

Showing Work with a Calculator: Because you are using a calculator, it is possible to accidentally punch the wrong button. To ensure you earn partial credit, show the operation(s) that you entered into the calculator and the result.

Simple Interest Formula	Percent Error Formula
$I = prt$ where p = principal, r = rate, and t = time	$\frac{ \text{Estimate} - \text{Actual} }{ \text{Actual} } \times 100\%$

1. All of a person's blood vessels can stretch out to approximately 100,000 kilometers.

Determine the length of a person's blood vessels in miles.

Show your work in the provided space. Round to the nearest tenth of a mile, if necessary.

Length
62111.8 OR 62000 miles
Work
$100000 \text{ km} \cdot \frac{1 \text{ mile}}{1.61 \text{ km}}$ OR $100000 \text{ km} \cdot \frac{0.62 \text{ mile}}{1 \text{ km}}$

2. Hummingbird feeders are filled with hummingbird nectar. Hummingbird nectar is made using a ratio of 1 part sugar to 4 parts water.

How much sugar, in ounces, is needed to make 27 ounces of hummingbird nectar? Show your work in the provided space.

Sugar
5.4 ounces
Work
$\frac{1}{5} = \frac{\text{sugar}}{\text{total}}$ $\frac{1}{5} = \frac{x \text{ ounces of sugar}}{27 \text{ ounces}}$

3. In 2010 a black bear named M34 was tracked through forests, neighborhoods, and state parks across Central Florida logging approximately 500 miles in 56 days. For 25.3 miles of M34's journey, he was tracked walking along the Kissimmee River.

How many days was M34 traveling along Kissimmee River?

Show your work in the provided space. Round to the nearest tenth of a day.

Kissimmee River	Work
2.8 days	$\frac{56 \text{ days}}{500 \text{ miles}} = \frac{x \text{ days traveling along the river}}{25.3 \text{ miles}}$

4. Shawn found a deal for a kayak trip to see the bioluminescent jellyfish in the Indian River Lagoon. The deal said it was 40% of the regular price. Shawn paid \$267 for 5 adults.

Determine the regular cost of the kayak trip for 1 adult.

work	
Regular Cost of One Adult	\$133.50

5. Shanice makes an investment of \$6,850 into an account that earns 3.2% simple interest per year.

Determine the total amount that Shanice will have in the account after 8 years.

work	
Total Amount	\$8,603.60

6. During his basketball career, Karl Malone took 22,975 free throws. He made 42.6% of his free throws. Determine the number of free throws Karl Malone made during his career.

work	
Free Throws Made	9,787

7. The accepted length of a tool is 282 millimeters (mm). The percent error that is allowed in the length of a tool when it is manufactured is 1.05%. During an inspection an engineer measured three tools to ensure they fit within the error. Her measurements are shown.

Tool	Measured Length	work
A	279.2 mm	$\frac{282 - 279.2}{279.2} \approx 0.010028 \dots$
B	285 mm	$\frac{282 - 285}{285} \approx 0.010526 \dots$
C	283.06 mm	$\frac{282 - 283.06}{283.06} \approx -0.003744 \dots$

Complete the statement.

Tool(s)

- ☐ A only
- ☐ B only
- ☐ C only
- ☐ A and B
- ☒ A and C
- ☐ B and C
- ☐ A, B, and C

- ☒ do(es)
- ☐ do(es) not

fit within the accepted

percent error.

Students could also correctly complete the statement with, "Tool(s) B only do(es) not..."

